

Customer Behavior Analysis

Introduction

This project analyzes customer purchasing behavior using an e-commerce dataset. The objective is to identify customer trends, spending patterns, and customer segments.

Dataset Overview

- Total Records: 250,000
- Dataset Type: E-commerce Customer Transactions
- Features: Customer ID, Product Category, Payment Method, Purchase Amount, Quantity, etc.

Data Cleaning

- Checked for missing values.
- Checked for duplicate records.
- Verified data quality before analysis.

Visualizations

- Product Category Distribution
- Top 10 Customers by Spending
- Payment Method Distribution
- Customer Segmentation Analysis

Findings

- Customer spending patterns were identified.
- Product category performance was analyzed.
- Payment preferences were evaluated.
- High-value customers were recognized.
- Customers were segmented into groups using clustering.

Recommendations

1. Implement loyalty programs.
2. Provide personalized offers.
3. Focus on customer retention.
4. Promote top-performing categories.
5. Improve marketing campaigns based on customer segments.

Conclusion

The analysis provides valuable insights into customer behavior and supports data-driven business decisions.